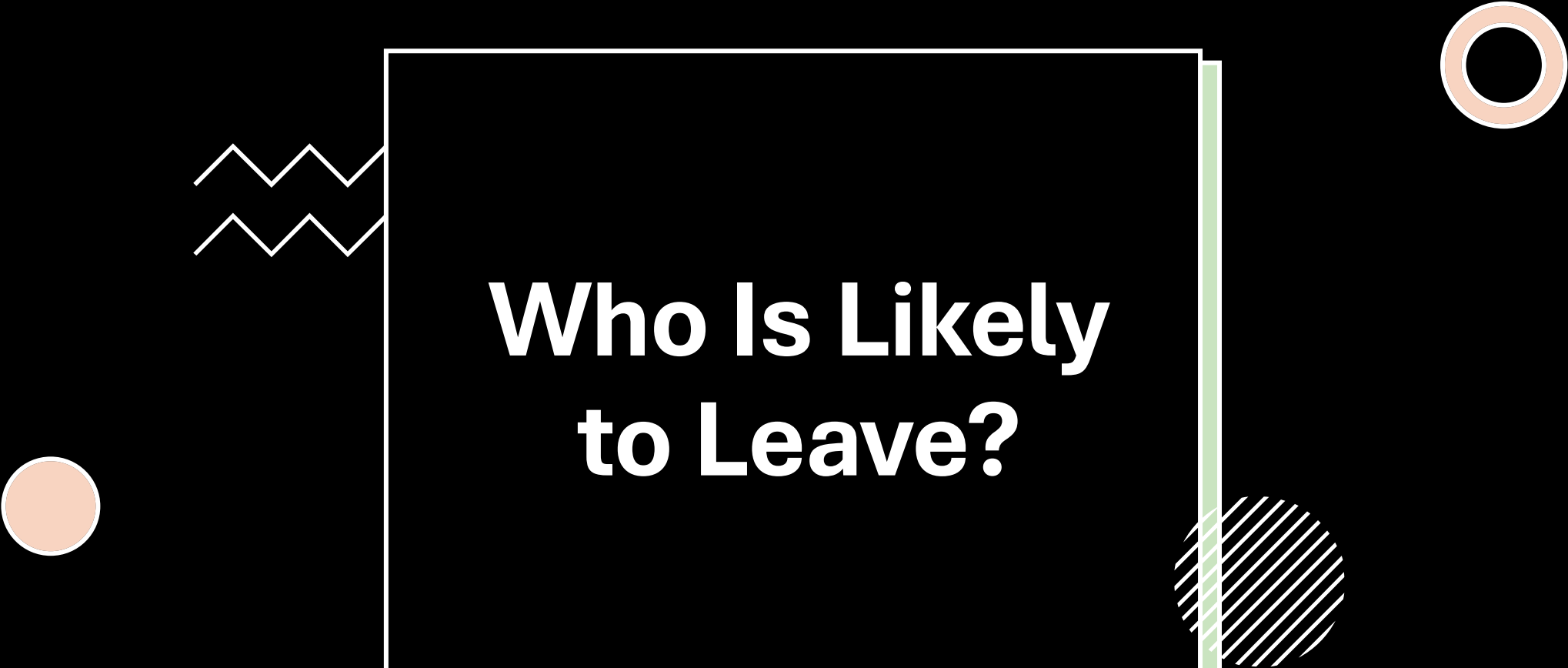




Who Is Likely to Leave?

- Predicting Customer Churn in a Telecom Company

Business Context

Why Customer Churn Matters



- Highly competitive telecom market
- Stable customer acquisition but stagnant revenue
- Customer churn reduces long-term profitability
- Retaining customers is cheaper than acquiring new ones

Business Objective

What Management Needs to Know



IDENTIFY CUSTOMERS
AT HIGH RISK OF
CHURN



UNDERSTAND KEY
DRIVERS OF CHURN
BEHAVIOR



SUPPORT TARGETED
RETENTION STRATEGIES



IMPROVE CUSTOMER
LIFETIME VALUE

Data Overview

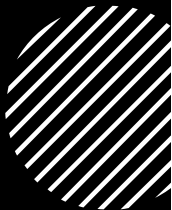
Telco Customer Churn Dataset

-
- Source: Kaggle Telco Customer Churn dataset
 - 7,043 individual customer records
 - Unit of analysis: customer account
 - Target variable: Churn (Yes / No)
 - Variables include demographics, services, and billing information



Exploratory Insights

Key Patterns in Customer Churn



26.5% of customers have churned



Churn is concentrated among early-tenure customers



Month-to-month contracts show higher churn



Fiber-optic customers have elevated churn risk



Modeling Approach

How Churn Was Predicted



Binary logistic regression model



Predicts probability of customer churn



Chosen for interpretability and transparency



Key predictors: tenure, contract type, internet service, charges



Key Model Results

Main Drivers of Customer Churn



Longer contracts significantly reduce churn risk



Two-year contracts reduce churn by approximately 85%



Fiber-optic customers are about three times more likely to churn



Longer customer tenure lowers churn probability

Model Performance

How Well Does the Model Work?

- Accuracy: approximately 77.6%
- AUC: approximately 0.82
- High specificity at default threshold
- Threshold tuning increases sensitivity to over 80%

Managerial Recommendations

Turning Insights into Action

01

Encourage month-to-month customers to adopt longer contracts

02

Focus retention efforts on early-tenure customers

03

Provide proactive support to fiber-optic customers

04

Use churn probabilities for targeted retention campaigns

Conclusion

Business Value of Predictive Analytics



- Identifies customers most likely to churn
- Supports proactive and data-driven retention strategies
- Improves allocation of retention resources
- Enhances long-term profitability

Thank You